# Research Framework Template

**IMMU-201 | Phase II Immunotherapy Trial**

*Synthetic training data - educational use only.*

## Instructions

Use the Protocol Summary Document as your source. Complete each section using study-specific evidence. Do not answer from memory. One section is modeled to demonstrate the expected tone, structure, and level of detail.

- Identify and annotate the primary research question.
- Identify the study hypothesis and explain how it relates to the research question.
- Identify the primary and secondary objectives.
- Classify the study approach and support the classification with evidence from the protocol summary.
- Explain briefly how the study design supports the scientific question.

## Section 1: Primary Research Question Annotation - MODELED EXAMPLE

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| **Prompt** | Identify the primary research question and annotate its key elements. |
| **Protocol Evidence** | Among adults with moderate-to-severe rheumatoid arthritis who remain symptomatic despite stable standard therapy, does IMT-201 improve disease activity at Week 24 compared with placebo? |
| **Annotation** | Population: adults with moderate-to-severe rheumatoid arthritis; Intervention: IMT-201; Comparator: placebo; Outcome: change in disease activity; Timeframe: Week 24. |
| **Why This Matters** | The question clearly identifies who is being studied, what intervention is being tested, what it is compared with, the outcome of interest, and when the primary comparison occurs. |

**Use the modeled section above as a reference.** Your remaining responses should be equally specific and should connect directly to language in the Protocol Summary.

## Section 2: Secondary Research Question(s)

**Prompt:** Identify the secondary research question(s) stated or supported by the protocol summary.

The Protocol Summary (Section 2) states two secondary research questions:

(1) "Does IMT-201 improve physical function and participant-reported pain?"

(2) "What safety and tolerability findings occur during treatment?"

**Study-specific response:**

**Secondary question 1 (efficacy beyond disease activity):** Population: adults with moderate-to-severe rheumatoid arthritis who remain symptomatic despite stable standard therapy; Intervention: IMT-201 subcutaneous injection every 4 weeks through Week 20; Comparator: matched placebo; Outcomes: physical function (validated participant questionnaire score) and participant-reported pain (0–10 numeric rating scale); Timeframe: baseline to Week 24.

**Secondary question 2 (safety and tolerability):** Population and arms as above; Outcomes: adverse events, serious adverse events, injection-site reactions, clinically significant laboratory abnormalities, and treatment discontinuations; Timeframe: during the 24-week treatment period.

Together these questions ask whether any improvement in disease activity is accompanied by benefits the participant can feel (function and pain) and whether IMT-201 is safe and tolerable enough to justify further development.

**Protocol evidence/support:**

Section 2 (Research Questions and Hypothesis) states both secondary questions. Section 3 turns them into three secondary objectives: change in physical function from baseline to Week 24, change in participant-reported pain from baseline to Week 24, and characterization of the safety and tolerability of IMT-201.

Section 7 supplies the matching endpoints: change from baseline to Week 24 in a validated participant physical-function questionnaire score; change from baseline to Week 24 in participant-reported pain on a 0–10 numeric rating scale; and incidence and severity of treatment-emergent adverse events and serious adverse events through Week 24.

## Section 3: Hypothesis Identification

**Prompt:** State the study hypothesis in your own words while preserving the meaning of the protocol.

Protocol wording (Section 2): "Participants randomized to IMT-201 will demonstrate a greater reduction in rheumatoid arthritis disease activity at Week 24 than participants randomized to placebo."

**Hypothesis:**

Among adults with moderate-to-severe rheumatoid arthritis who are still symptomatic on stable standard therapy, those assigned to IMT-201 will show a larger decrease in disease activity by Week 24 than those assigned to matched placebo.

This is a directional superiority hypothesis: it predicts both that the groups will differ and which group will do better.

**Connection to the primary research question:** Explain how the hypothesis predicts an answer to the primary research question.

The primary research question asks whether IMT-201 improves disease activity at Week 24 compared with placebo. The hypothesis gives the predicted answer: yes, with a greater reduction in the IMT-201 arm.

The hypothesis uses the same population (adults with moderate-to-severe rheumatoid arthritis), intervention (IMT-201), comparator (placebo), outcome (disease activity), and timepoint (Week 24) as the question. It is tested by the primary endpoint in Section 7, the change from baseline to Week 24 in the protocol-defined rheumatoid arthritis disease activity score.

## Section 4: Study Objectives

| Objective Type | Objective | How It Connects to the Study Question |
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| Primary Objective | To evaluate the efficacy of IMT-201 compared with placebo in reducing disease activity from baseline to Week 24. (Protocol Summary, Section 3) | This objective restates the primary research question as the task the study must accomplish: the same intervention (IMT-201), comparator (placebo), outcome (disease activity), and timeframe (baseline to Week 24). It is the objective the hypothesis is tested against and is measured by the primary endpoint, the change from baseline to Week 24 in the protocol-defined disease activity score. |
| Secondary Objective(s) | 1. To evaluate change in participant physical function from baseline to Week 24.<br><br>2. To evaluate change in participant-reported pain from baseline to Week 24.<br><br>3. To characterize the safety and tolerability of IMT-201, including adverse events, serious adverse events, | Objectives 1 and 2 answer the first secondary research question ("Does IMT-201 improve physical function and participant-reported pain?"). They show whether a change in disease activity is accompanied by improvement in how participants function and feel. |

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| | injection-site reactions, clinically significant laboratory abnormalities, and treatment discontinuations.<br><br>(Protocol Summary, Section 3) | Objective 3 answers the second secondary research question ("What safety and tolerability findings occur during treatment?"). It provides the risk side of the benefit-risk picture, which is essential for an investigational therapy that alters immune activity. |

## Section 5: Study Approach Classification

**Prompt:** Classify the study approach (for example, interventional or observational) and support your classification using evidence from the Protocol Summary.

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| **Study Approach Classification** | Interventional study: a Phase II, multicenter, randomized, double-blind, placebo-controlled, parallel-group clinical trial. |
| **Protocol Evidence** | Section 1 lists "Study Approach: Interventional" and the study title is "A Phase II, Randomized, Double-Blind, Placebo-Controlled Study of IMT-201 Immunotherapy in Adults with Moderate-to-Severe Rheumatoid Arthritis."<br><br>Section 4 states that "Approximately 120 eligible participants will be randomized 1:1 to IMT-201 or matched placebo" and that study treatment "will be administered by subcutaneous injection every 4 weeks through Week 20." |
| **Reasoning** | The study is interventional because the protocol, not routine clinical care, determines what each participant receives: participants are assigned by randomization to an investigational product or placebo and are then followed prospectively on a fixed visit schedule to measure the effect. An observational study would only record outcomes of treatment chosen in ordinary practice, with no assignment, placebo, or blinding. |
| **Key Design Features** | **Randomization:** approximately 120 participants randomized 1:1 to IMT-201 or matched placebo.<br><br>**Blinding:** double-blind; participants, investigators, coordinators involved in outcome assessment, and designated study personnel remain blinded unless emergency unblinding is |

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| | medically necessary.<br><br>**Control group:** matched placebo injection on the same schedule, added to stable protocol-permitted background therapy.<br><br>**Parallel-group design:** each participant remains in the assigned arm for the 24-week treatment period.<br><br>**Setting and phase:** Phase II, conducted at multi-site academic medical centers.<br><br>**Schedule:** injections every 4 weeks through Week 20; assessments at Screening, Baseline/Day 1, and Weeks 4, 8, 12, 16, 20, and 24. |

## Section 6: Design-to-Question Connection

**Prompt:** Explain how the study design supports answering the primary scientific question.

Scientific question being answered: among adults with moderate-to-severe rheumatoid arthritis who remain symptomatic despite stable standard therapy, does IMT-201 improve disease activity at Week 24 compared with placebo?

**Response:**

The question is comparative and causal, so the design has to isolate the effect of IMT-201 from everything else that could change disease activity over 24 weeks. Each design feature removes one competing explanation:

**Randomization (1:1):** assigns participants to IMT-201 or placebo by chance, so the two arms are expected to be similar at baseline and a difference at Week 24 can be attributed to the treatment rather than to who received it.

**Matched placebo control on the same injection schedule:** shows what happens without IMT-201, including natural fluctuation in rheumatoid arthritis and the response to receiving injections and study attention.

**Double-blinding:** participants, investigators, and coordinators involved in outcome assessment do not know the assignment. This matters because disease activity examinations involve clinical judgment and the pain and function measures are participant-reported.

**Parallel groups with stable background therapy:** each participant stays in one arm for the full 24 weeks and continues stable, protocol-permitted background therapy, so the comparison is IMT-201 plus standard therapy versus placebo plus standard therapy, which matches the "remain symptomatic despite stable standard therapy" population in the question.

**Assessment schedule:** disease activity is measured at Baseline/Day 1 and again at Week 24, the two timepoints needed to calculate the change the question asks about, with visits every 4 weeks in between for dosing and safety monitoring.

**Phase II scale:** approximately 120 participants across multiple academic sites is appropriate for an initial controlled estimate of efficacy and safety in the target population.

**Evidence from the protocol summary:**

Section 4: "Phase II, multicenter, randomized, double-blind, placebo-controlled, parallel-group clinical trial"; "Approximately 120 eligible participants will be randomized 1:1 to IMT-201 or matched placebo"; participants, investigators, and coordinators involved in outcome assessment "will remain blinded to treatment assignment unless emergency unblinding is medically necessary"; dosing "every 4 weeks through Week 20, with the primary endpoint assessed at Week 24"; participants "continue protocol-permitted stable background rheumatoid arthritis therapy"; assessments at Screening, Baseline/Day 1, and Weeks 4, 8, 12, 16, 20, and 24.

Section 6: placebo is a "matched placebo" given by "subcutaneous injection on the same schedule." Section 7: the primary endpoint is the "change from baseline to Week 24 in the protocol-defined rheumatoid arthritis disease activity score."

# Final Self-Check

☑️ I used the Protocol Summary Document rather than relying on memory.

☑️ My research question annotation identifies the study-specific population, intervention, comparator, outcome, and timeframe.

☑️ My hypothesis is consistent with the study's primary research question.

☑️ I identified the primary and secondary objectives accurately.

☑️ I classified the study approach and supported it with protocol evidence.

☑️ My explanation connects the study design to the scientific question.

☑️ I replaced all blank response areas with complete, study-specific answers before submission.

**Training note:** The study and all protocol information used in this template are synthetic and were created solely for this learning activity.